Mid01 Updates Rules 10/22

1. 禁止後處理與 Leaderboard-based Fitting

正式評分所使用的結果必須由各組的 Localization Pipeline 直接產生。禁止在完成 Localization 後，再針對軌跡做額外的離線修正，例如：① 對整段軌跡進行平移或旋轉校正，② 根據完整軌跡進行非因果平滑，③ 人工調整部分或全部 Pose，④ 其他不屬於 Localization Pipeline 正常執行流程的後處理

此外，一般的演算法開發、模型比較、參數調整與實驗仍然允許，但嚴禁利用 Kaggle Public Leaderboard 的分數回饋，透過多次提交、系統性修改或擾動預測結果反推出 GT，直接擬合最終預測結果。

2. Causal / Online Localization

本次競賽要求 Localization 演算法遵守 Causal Processing / Online Localization 原則。在估計當前 Frame 的 Pose 時，只能使用當前及過去 Frame 的資訊，不得使用未來 Frame 的資訊回頭修正先前的定位結果。

例如，在估計 Frame 300 的 Pose 時，可以使用 Frame 1–300 的資訊；但在取得 Frame 301–500 的資料後，不得再回頭修改 Frame 300 的 Pose。

允許的方式包括：

- 使用過去的 Pose 或 Sensor Measurements
- Kalman Filter / Extended Kalman Filter
- Moving Average 或其他 causal filtering 方法
- 其他僅使用當前與歷史資訊的Localization 方法

禁止的方式包括：

- 使用未來 Frame 修正過去的 Pose
- Backward Smoothing
- Non-causal Smoothing
- 完成整段 Localization 後，再根據完整軌跡進行離線修正
- 任何利用完整測試序列資訊回頭調整先前結果的方法

3. Kaggle 最終提交與程式驗證

請於 10/29 9:00 前，在 Kaggle 選取符合規範、由實際 Localization Pipeline 產生的 CSV 作為最終評分 Submission。

競賽結束後，助教將檢 查各組繳交的程式碼，並重新執行 Localization Pipeline，確認最終結果是否可以重現。
重新執行所 產生的結果應與各組最終選定的 Kaggle Submission 基本一致，僅允許合理範圍 內因浮點運算、套件版本或硬體差異所造成的數值誤差。若重新執行結果與最終 Submission 存在明顯差異，我們將進一步檢 查：程式碼、執行環境、使用的參數與設定、資料處理流程、最終 Submission 的產生方式…

若經確認存在以下情況之一，該 Track 將以 0 分計算：

- 違規的離線軌跡後處理
- Leaderboard-based Fitting
- 利用 Kaggle 分數反推或修正 Ground Truth / Localization Pose
- 最終 Submission 無法由所繳交的程式碼與設定合理重現
- 其他刻意規避競賽規範的行為

4. 程式碼繳交要求

請各組在繳交前務必確認：

① 程式碼符合 Causal / Online Localization 原則。
② 不包含違規的離線軌跡後處理。
③ 不包含利用 Kaggle Leaderboard feedback 擬合最終結果的流程。
④ 所繳交的程式碼、參數及設定可以重現最終選定的 Kaggle Submission。
⑤ README 中提供完整的執行流程、必要環境與參數設定。

請務必確認自己的方法與提交 內容符合規範。

1. Prohibition of Post-processing and Leaderboard-based Fitting

All results used for official evaluation **must be generated directly by each team's localization pipeline**.
Additional offline trajectory corrections after localization are strictly **prohibited**, including:
1. Applying translation or rotation corrections to the entire trajectory.
2. Performing non-causal smoothing using the complete trajectory.
3. Manually adjusting some or all estimated poses.
4. Any other post-processing outside the normal execution of the localization pipeline.

Standard algorithm development, model comparison, parameter tuning, and experimentation are still allowed.

However, **using Kaggle Public Leaderboard feedback to infer hidden ground truth** through repeated submissions, systematic modifications, or perturbations of predicted results, and then directly fitting the final predictions, is strictly **prohibited**.

2. Causal / Online Localization

All localization algorithms must follow the principles of **Causal Processing / Online Localization**.

When estimating the pose at the current frame, **only information from the current and previous frames may be used**. Future frame information must not be used to retrospectively modify previously estimated poses.

For example, when estimating the pose at Frame 300, information from Frames 1–300 may be used. However, after obtaining data from Frames 301–500, the estimated pose at Frame 300 must not be modified.

<u>Allowed Methods</u>
- Using previous poses or sensor measurements.
- Kalman Filter / Extended Kalman Filter.
- Moving Average or other causal filtering methods.
- Other localization methods that use only current and historical information.

<u>Prohibited Methods</u>
- Using future frames to correct past poses.
- Backward Smoothing.
- Non-causal Smoothing.
- Applying offline trajectory corrections after completing localization.
- Any method that uses information from the complete test sequence to retrospectively modify earlier results.

3. Final Kaggle Submission and Code Verification

Before October 29 at 9:00 AM, each team must select a final Kaggle submission (CSV) that complies with the competition rules and is generated directly by their localization pipeline.

After the competition, the TAs will review the submitted code and rerun the localization pipeline to verify the reproducibility of the final results. The reproduced results must be consistent with the selected Kaggle submission, allowing only reasonable numerical differences caused by floating-point computations, software versions, or hardware differences.
If significant discrepancies are found, we will further examine the source code, execution environment, parameters, configurations, data processing pipeline, and submission generation process.

If any of the following violations are confirmed, the corresponding Track will receive a score of zero:

- Prohibited offline trajectory post-processing.
- Leaderboard-based fitting.
- Using Kaggle scores to infer hidden ground truth or correct localization poses.
- Final submissions that cannot be reasonably reproduced using the submitted code and configurations.
- Any other deliberate attempts to circumvent the competition rules.

4. Code Submission Requirements

Before submitting your code, each team must ensure that:

1. The code follows Causal / Online Localization principles.
2. No prohibited offline trajectory post-processing is included.
3. No Kaggle leaderboard feedback is used to fit or correct the final results.
4. The submitted code, parameters, and configurations can reproduce the selected final Kaggle submission.
5. The README provides complete execution instructions, required dependencies, and parameter configurations.

Please carefully verify that your methods, code, and final submissions comply with all competition rules.